Remember the reason.

Final product report for the verified Round 3 implementation

DeadEnd preserves the technical approaches a team rejected, the evidence behind those decisions, and the conditions an incoming developer must recheck.

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Working public demo: deadend-ai-codersinpune.vercel.app
Signed-in team workspace: deadend-ai-codersinpune.vercel.app/team.html
Public repository: github.com/nikkushrivastava0-a11y/deadend-ai

What a judge can do without credentials

Open the fictional Stripe case. Inspect its exact rejection quote and possible later update. Correct the interpretation, approve a public finding and download the Markdown handover. The public demo is separate from signed-in team data.

What a judge can do with their own accounts

Open Team workspace and create a unique lowercase handle plus a password of at least 12 characters. Create a workspace as owner and load the fictional case. Create a one-use recipient invitation. A second account joins using that code. The owner approves and issues a brief; the recipient downloads and acknowledges it. Reopen or sign in again to inspect the stored receipt.

Release evidence

Implementation merged through PR #13; functional source release fe6a32b. The local build and 38 automated tests passed, including the JSONB persistence regression found in hosted testing. Hosted preview account flows passed. Final production verification is recorded in PROGRESS.md and the release notes; this PDF must accompany that verified release, not an earlier Round 2 build.

The three promises delivered.

Exact commitments from the submitted Round 2 report

01 / Decision revalidation

Owners record a dated current-constraint assessment: unknown, still-blocked or changed. Changed/still-blocked status requires a source-bound supporting excerpt. Every original quotation is retained, history is append-only, invalid/future dates are refused, and a new assessment reopens review. The handover carries the latest dated assessment and its uncertainty.

02 / Cross-source disagreement resolver

A reviewer explicitly compares a finding with an excerpt from another authorised source. Both excerpts stay verbatim. A human records whether the original applies, the second applies, context matters, or the issue remains unresolved, with rationale and next action. An unresolved disagreement blocks approval/export; a resolution still needs fresh approval.

03 / Role-aware handover and receipt

Individual application accounts have workspace-specific owner, reviewer or recipient membership. Owners invite by one-use, expiring codes. The server filters restricted sources and other recipients' briefs. Approved public evidence is frozen into a versioned brief. The addressed recipient can download and acknowledge it. Duplicate issue/receipt actions preserve the original record, and shared PostgreSQL state survives relogin and function restarts.

Preserved evidence boundary

Only approved public findings enter an issued brief. Restricted notes remain owner-only. Verifying that a quote occurs in a source establishes provenance, not that the technical assertion is true today. Revalidation and reconciliation are dated human judgments, not automated declarations of current truth.

Working feature inventory.

Implemented behaviour and architecture

Completed working features

Selected public GitHub issue/PR import, including discussion and review segments; permitted note import; deterministic rejection extraction; exact-quote/comment-link checks; human interpretation correction; approval/rejection/dispute/reopen; related-update leads; approved Markdown export; genuine desktop Three.js evidence atlas with semantic fallback; synthetic credential-free demo; dated recheck history; explicit two-source resolution; individual team accounts; invitations; role-aware source/brief visibility; shared revisioned storage; immutable approved briefs; recipient downloads and receipts.

Architecture

Frontend: vanilla HTML/CSS/JavaScript and self-hosted Three.js/assets. The anonymous case room and signed-in Team workspace are separate interfaces.
Backend: existing Node.js HTTP server, reused by the Vercel API adapter. Domain workflows share the same evidence/review functions.
Storage: dedicated free-quoted Supabase PostgreSQL project in Mumbai. JSONB workspace snapshots use optimistic version checks. Private tables have RLS and no direct client table privileges.
Authentication: custom salted scrypt accounts and random eight-hour HttpOnly/SameSite session cookies, Secure on HTTPS. Session tokens are hashed in storage and revoked by logout. This is not Supabase Auth.
Database access: a scoped server-only capability gates a fixed RPC. A public SECURITY INVOKER wrapper delegates to a private function with a locked search path. No service-role key is used or shipped to the client.

Role matrix

Owner: authorised source import, all workspace sources, dated rechecks, human review, invitations and public brief issuance.
Reviewer: public evidence, corrections/reviews and reconciliation; no restricted sources or brief issuance.
Recipient: only assigned immutable briefs and their own receipt action. Roles are looked up from current database membership; a browser cannot choose its own role.

Verification that mattered.

Observed tests; no invented benchmark or usability score

Automated build and regression checks

npm run check passed 38/38 tests. Coverage includes source import/provenance, invented or missing quotations, restricted output, model outage fixtures, source API failure, session tampering, review correction, dated rechecks, stale writes, two-source resolution, roles, other-recipient isolation, immutable briefs and restart recovery. GitHub CI passed on the final reviewed candidate.

Actual hosted account journey

Separate disposable owner, reviewer, recipient and outsider accounts exercised registration/login; one-use invitations; workspace isolation; restricted-source filtering; owner-only revalidation; two-source human reconciliation; mandatory fresh approval; frozen brief content after later rejection; duplicate issue/receipt; and receipt recovery after logout/relogin. Public production checks use fresh clients and synthetic data only.

A defect found and fixed

PostgreSQL JSONB reorders object keys. The initial JSON.stringify-based comparison treated the same approved content as different and issued another brief version. Deep structural equality fixes this. A dedicated regression round-trips reordered keys and asserts the original brief ID and single-version count. The revised hosted journey passed this case.

Browser evidence and limits

The Team sign-in page was inspected in the cloud browser at 1363px with no document overflow. Earlier public demo review-to-handover browser checks covered desktop/mobile emulation. This release does not claim a full signed-in browser E2E or physical-phone test. Signed-in multi-account behaviour was verified via real hosted HTTP API requests, not a visual mock.

Run it independently

Clone the public repository; use Node 22+; run npm ci and npm run check, then npm start. The anonymous workflow runs without a database. To reproduce hosted team mode, apply database/hosted-team.sql to a dedicated project and configure the three server-only variables documented in docs/HOSTED_TEAM.md. No credentials are committed.

Limits and next development.

Honest scope, roadmap, contributions and submission

Known limitations

No email verification, password recovery, OAuth or MFA. Handles are application identities, not verified real-world names. Use a unique strong password. No private GitHub token import or live model credential is configured. Production extraction is deterministic; optional structured model handling was fixture-tested. Imports are bounded by GitHub limits and explicit rejection language. A closed PR alone is not proof an approach failed. Only approved public evidence can be issued; owner-only restricted notes cannot be privately handed over. No claim of a ten-page enterprise product, perfect semantic contradiction detection, physical-mobile QA or measured performance score is made.

Changes from the Round 1/2 implementation

The original concept remains recovering negative project knowledge. The implementation retained the working Node/vanilla/Three.js architecture rather than restarting in Next.js. Round 3 added dated human revalidation, explicit two-source reconciliation and real shared account/receipt storage. Anonymous encrypted browser sessions remain available for credential-free judging; they are not used as cross-user authorisation.

Roadmap

1. Add verified identity, account recovery, membership revocation UI and granular source grants, with migration/recovery tests.
2. Add authorised private GitHub integration and incremental source refresh while preserving original evidence versions.
3. Evaluate live structured extraction on labelled real discussions, measure error/abstention rates and improve semantic retrieval without weakening exact-quote checks.

Team and AI assistance

Nitanshu Shrivastav leads product direction, integration and release oversight. Backend/storage implementation, tests and documentation used AI assistance. Kanishk Baghel is the registered teammate assigned frontend/mobile polish and independent QA; unverified separate work is not counted as completed in this report.

Submission

Team leader submits once by 16 November 2026, 23:59 IST: final app URL, repository, this product report, feature list, architecture, improvements, testing notes and roadmap. No video or social-media submission is required. The final form has not been submitted automatically.